

Qualifying Function of One Complex Variable I

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Chapter 1

August 2018

1.1 Part I

1. *Find a conformal map f from the vertical strip*

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk $\mathbb{D}$.

Solution. The width of the strip is $11 - 3 = 8$. First map the strip to a horizontal strip by setting

$$w = \frac{\pi}{8}(z - 3).$$

Since $3 < \operatorname{Re}(z) < 11$, we have

$$0 < \operatorname{Re}(w) < \pi.$$

Now define

$$\omega = e^{iw}.$$

Because

$$\arg(\omega) = \operatorname{Re}(w),$$

the map $\omega = e^{iw}$ sends the strip onto the upper half-plane

$$\{\omega \in \mathbb{C} : \operatorname{Im}(\omega) > 0\}.$$

Finally, the map

$$\phi(\omega) = \frac{\omega - i}{\omega + i}$$

maps the upper half-plane conformally onto the unit disk $\mathbb{D}$. Therefore a conformal map from Ω onto $\mathbb{D}$ is

$$f(z) = \frac{e^{i\pi(z-3)/8} - i}{e^{i\pi(z-3)/8} + i}.$$

■

2. Suppose that F and G are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that F and G have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function h on H such that

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$.

Solution. If $F \equiv G \equiv 0$, then we may take $h \equiv 0$. Thus assume that F and G are not identically zero.

Define

$$q(z) = \frac{F(z)}{G(z)}$$

at points where $G(z) \neq 0$. We claim that q extends to a holomorphic function on all of H .

Let $a \in H$ be a zero of G . Since F and G have the same zeros with the same multiplicities, there is some $m \geq 1$ such that locally

$$F(z) = (z - a)^m \phi(z), \quad G(z) = (z - a)^m \psi(z),$$

where ϕ and ψ are holomorphic near a , with

$$\phi(a) \neq 0, \quad \psi(a) \neq 0.$$

Hence near a ,

$$q(z) = \frac{F(z)}{G(z)} = \frac{\phi(z)}{\psi(z)}.$$

So the singularity of q at a is removable. Therefore q extends holomorphically to all of H .

Moreover, the extension of q has no zeros. Indeed, away from the zeros of G , this is clear since F and G have the same zeros. At a common zero a , we have

$$q(a) = \frac{\phi(a)}{\psi(a)} \neq 0.$$

Now $H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$ is convex, hence simply connected. Since q is holomorphic and never zero on H , by Corollary 6.17, Chapter IV, there exists a holomorphic function h on H such that

$$q(z) = e^{h(z)}.$$

Therefore, for all $z \in H$,

$$F(z) = q(z)G(z) = e^{h(z)}G(z).$$

Thus there is a holomorphic function h on H satisfying

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$. ■

3. Suppose f is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence $(z_n)_{n=1}^\infty$ in D such that $|z_n| \rightarrow 1$ and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

Solution (I). If $f \equiv 0$, then take $z_n = 1 - \frac{1}{n}$. Then $|z_n| \rightarrow 1$ and $\{f(z_n) : n \geq 1\} = \{0\}$, which is bounded.

Now assume $f \not\equiv 0$. Suppose, toward a contradiction, that no such sequence

exists. Then for every $M > 0$, there is $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Indeed, otherwise we could find $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give $|z_n| \rightarrow 1$ and $\{f(z_n)\}$ bounded.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, for some $0 < R < 1$,

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence all zeros of f lie in the compact set $\{|z| \leq R\}$. Since $f \not\equiv 0$, it has only finitely many zeros there. Let those zeros be $\alpha_1, \dots, \alpha_N$, with multiplicities $m_1, \dots, m_N$, and define

$$P(z) = \prod_{j=1}^N (z - \alpha_j)^{m_j}.$$

If f has no zeros, take $P \equiv 1$.

Then

$$F(z) = \frac{f(z)}{P(z)}$$

extends to a holomorphic function on D , and F has no zeros in D . Therefore

$$g(z) = \frac{1}{F(z)}$$

is holomorphic on D .

Since P is bounded on D , and $|f(z)| \rightarrow \infty$ as $|z| \rightarrow 1$, we get

$$|F(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

Thus

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Fix $z_0 \in D$. Given $\varepsilon > 0$, choose $\rho < 1$ such that

$$|g(z)| < \varepsilon$$

whenever $\rho < |z| < 1$. Choose r such that

$$\max\{|z_0|, \rho\} < r < 1.$$

By the maximum modulus principle applied on $\{|z| \leq r\}$,

$$|g(z_0)| \leq \max_{|z|=r} |g(z)| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $g(z_0) = 0$. Hence $g \equiv 0$ on D , which is impossible because $g = 1/F$.

Therefore our supposition was false. Hence there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

Solution (II). Suppose, toward a contradiction, that no such sequence exists. Then for every $M > 0$, there exists $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Otherwise, for some $M > 0$, we could choose $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give the desired sequence.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, there exists $0 < R < 1$ such that

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence f has no zeros in the annulus

$$A = \{z : R < |z| < 1\}.$$

Therefore

$$g(z) = \frac{1}{f(z)}$$

is holomorphic on A . Also,

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Now expand g in its Laurent series on the annulus A :

$$g(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

For any r with $R < r < 1$,

$$c_n = \frac{1}{2\pi i} \int_{|\omega|=r} \frac{g(\omega)}{\omega^{n+1}} d\omega.$$

Hence

$$|c_n| \leq \max_{|\omega|=r} |g(\omega)| r^{-n}.$$

Letting $r \rightarrow 1^-$, we have

$$\max_{|\omega|=r} |g(\omega)| \rightarrow 0,$$

and $r^{-n} \rightarrow 1$. Therefore

$$c_n = 0$$

for every $n \in \mathbb{Z}$. Hence

$$g \equiv 0$$

on A , which is impossible since $g = 1/f$.

Therefore our supposition was false. Thus there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

4. Determine, with proof, all entire functions f such that for all $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and all $z \in \mathbb{C}$ the equation

$$f(\lambda z) = \lambda f(z)$$

holds.

Solution (I). We claim that the only such functions are

$$f(z) = cz$$

for some constant $c \in \mathbb{C}$.

Let f be entire. Then f has a power series expansion on all of $\mathbb{C}$:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

For $|\lambda| = 1$, we have

$$f(\lambda z) = \sum_{n=0}^{\infty} a_n (\lambda z)^n = \sum_{n=0}^{\infty} a_n \lambda^n z^n.$$

By assumption,

$$f(\lambda z) = \lambda f(z) = \lambda \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

Therefore, for every $|\lambda| = 1$,

$$\sum_{n=0}^{\infty} a_n \lambda^n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

By uniqueness of power series coefficients,

$$a_n \lambda^n = \lambda a_n$$

for every $n \geq 0$ and every $|\lambda| = 1$. Hence

$$a_n (\lambda^n - \lambda) = 0.$$

If $n = 1$, this condition gives no restriction on a_1 . But if $n \neq 1$, we can choose $|\lambda| = 1$ such that

$$\lambda^n \neq \lambda.$$

Thus $a_n = 0$ for every $n \neq 1$. Therefore

$$f(z) = a_1 z.$$

Conversely, every function of the form $f(z) = cz$ satisfies

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Hence the entire functions satisfying the condition are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

Solution (II). First put $z = 0$. Then

$$f(0) = \lambda f(0)$$

for every $|\lambda| = 1$. Choosing $\lambda \neq 1$, we get

$$f(0) = 0.$$

Hence

$$g(z) = \frac{f(z)}{z}$$

is holomorphic on $\mathbb{C} \setminus \{0\}$, and the singularity at 0 is removable. Define

$$g(0) = f'(0).$$

Then g is entire.

For $z \neq 0$ and $|\lambda| = 1$,

$$g(\lambda z) = \frac{f(\lambda z)}{\lambda z} = \frac{\lambda f(z)}{\lambda z} = \frac{f(z)}{z} = g(z).$$

Thus, by continuity,

$$g(\lambda z) = g(z)$$

for all $z \in \mathbb{C}$ and all $|\lambda| = 1$.

In particular, if $|\zeta| = 1$, then $\zeta = \lambda \cdot 1$ for some $|\lambda| = 1$, so

$$g(\zeta) = g(1).$$

Thus g is constant on the unit circle.

Write the Taylor expansion of g at 0:

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

By Cauchy's integral formula for Taylor coefficients,

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{g(\zeta)}{\zeta^{n+1}} d\zeta.$$

For $n \geq 1$, since $g(\zeta) = g(1)$ on $|\zeta| = 1$, we get

$$a_n = \frac{g(1)}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta^{n+1}} d\zeta = 0.$$

Therefore all Taylor coefficients of g except possibly a_0 vanish. Hence g is constant on $\mathbb{C}$. Say

$$g(z) = c.$$

Then

$$f(z) = zg(z) = cz.$$

Conversely, if $f(z) = cz$, then

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Therefore the desired functions are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

5. Let f and g be entire functions satisfying $f(0) = g(0) \neq 0$ and

$$|f(z)| \leq |g(z)| \quad \forall z \in \mathbb{C}.$$

Show that $f = g$.

Solution. Since $g(0) = f(0) \neq 0$, there exists $r > 0$ such that

$$g(z) \neq 0$$

for all $z \in B(0, r)$. Define

$$h(z) = \frac{f(z)}{g(z)}$$

on $B(0, r)$. Then h is holomorphic on $B(0, r)$. Also, by hypothesis,

$$|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$$

for all $z \in B(0, r)$. Moreover,

$$h(0) = \frac{f(0)}{g(0)} = 1.$$

Thus

$$|h(0)| = 1.$$

So h attains its maximum modulus at the interior point 0. By the maximum modulus principle, h is constant on $B(0, r)$. Since $h(0) = 1$, we get

$$h \equiv 1$$

on $B(0, r)$. Therefore

$$f(z) = g(z)$$

for all $z \in B(0, r)$.

Since f and g are entire and agree on a nonempty open set, the identity theorem implies that

$$f = g$$

on all of $\mathbb{C}$.

■

1.2 Part II

6. Determine all entire functions that satisfy the functional equation

$$f(3z) = (1 - 3z)f(z).$$

Express the answer in terms of an infinite product.

7. Let $F(z)$ be holomorphic on a disc $D(z_0, r)$ so that

$$F(z_0) = F'(z_0) = 0 \quad \text{and} \quad F''(z_0) \neq 0.$$

Show that there are two curves Γ_1 and Γ_2 passing through z_0 that are orthogonal at z_0 and so that the function F when restricted to Γ_1 is real-valued and has a minimum at z_0 , and when restricted to Γ_2 it is also real-valued and has a maximum at z_0 .

8. Prove that if f is a non-constant bounded holomorphic function defined on the half plane

$$\{z : \operatorname{Im} z < 1\},$$

then there is some point on the real axis where the value of f is not real.

9. Show that there exists a sequence of polynomials (P_n) so that

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \text{if } \operatorname{Re}(z) > 0, \\ 0, & \text{if } \operatorname{Re}(z) = 0, \\ -1, & \text{if } \operatorname{Re}(z) < 0. \end{cases}$$

Also, explain why (P_n) can't be bounded on compact subsets of $\mathbb{C}$.

10. Let $\mathfrak{F}$ be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that $\mathfrak{F}$ is a normal family.

August 2020

2.1 Part I

2.2 Part II

Chapter 3

August 2024

3.1 Part I

3.2 Part II

Chapter 4

January 2025

4.1 Part I

4.2 Part II